

4. Linear Discriminant Analysis

For the linear discriminant analysis, the dataset containing 120 observations and 6 variables was imported. Variables X1 to X5 correspond to the concentrations of atmospheric pollutants and were used as explanatory variables, since they are quantitative variables. Variable X6 corresponds to the year of observation, assuming the categories A2013, A2014 and A2015, and was used as the grouping variable.

Initially, the data were organized by separating the quantitative variables from the qualitative variable. Therefore, variables X1, X2, X3, X4 and X5 were used to discriminate the three groups defined by variable X6.

The linear discriminant analysis was performed with the objective of assessing whether the concentrations of the five atmospheric pollutants allow the observations corresponding to the years 2013, 2014 and 2015 to be distinguished. Thus, the aim was to verify whether the atmospheric pollution profile varies sufficiently between the three years to allow the classification of the observations.

4.1. Characterization of the groups and verification of the assumptions

The grouping variable considered was the year of observation. The sample is balanced, since each group contains 40 observations. Since the three groups have the same size, the a priori probabilities are equal for all groups and are given by $40/120 = 0.3333$. Therefore, each observation initially has the same probability of belonging to any of the three years considered.

Group	Number of observations	A priori probability
A2013	40	0.3333
A2014	40	0.3333
A2015	40	0.3333

Table 1 - Number of observations and a priori probabilities for each group.

```
Prior probabilities of groups:
      A2013      A2014      A2015
0.3333333 0.3333333 0.3333333
```

Figure 1 - R output showing the a priori probabilities of the three groups.

Before applying LDA, the main assumptions of the method were considered. Linear discriminant analysis assumes that the quantitative variables have an approximately multivariate normal distribution within each group and that the variance-covariance matrices are similar across the groups.

The multivariate normality assumption was assessed separately for each group. The results indicated departures from multivariate normality in the three groups. However, because all groups have equal sizes and contain 40 observations each, this violation does not necessarily prevent the use of LDA. The results should nevertheless be interpreted with some caution.

The homogeneity of the variance-covariance matrices was assessed using Box's M test, at a significance level of $\alpha = 0.05$. The hypotheses considered were:

H0: The variance-covariance matrices are equal across the groups.

H1: The variance-covariance matrices are not equal across the groups.

Box's M-test for Homogeneity of Covariance Matrices

data: X
Chi-Sq (approx.) = 506.13, df = 30, p-value < 2.2e-16

Table 2 - Box's M test for equality of variance-covariance matrices.

Decision: Since the p-value is lower than the significance level $\alpha = 0.05$, the null hypothesis is rejected. Therefore, there is statistical evidence that the variance-covariance matrices are not equal across the three groups. This means that one of the assumptions of LDA is not fully satisfied.

Nevertheless, the three groups have exactly the same size, with 40 observations in each group. For this reason, LDA can still be applied as an exploratory and classification method, but the results should be interpreted carefully. In particular, the classification results and the graphical representation of the discriminant functions should be analysed together, rather than relying only on the formal assumptions.

4.2. Group means

Before applying linear discriminant analysis, the means of the quantitative variables were analysed for each group in order to identify possible initial differences between the years.

Year	X1	X2	X3	X4	X5
A2013	16.25	12.35	30.28	532.50	56.80
A2014	129.57	35.98	86.12	2425.00	4.53
A2015	58.78	17.80	51.60	1102.50	30.38

Table 3 - Mean values of variables X1 to X5 by year.

```
> aggregate(data[, 1:5], by = list(Year = data$X6), mean)
  Year      X1      X2      X3      X4      X5
1 A2013 16.250 12.350 30.275 532.5 56.800
2 A2014 129.575 35.975 86.125 2425.0 4.525
3 A2015  58.775 17.800 51.600 1102.5 30.375
```

Figure 2 - R output with group means for variables X1 to X5.

From the analysis of Table 3, clear differences between the three years can be observed. The year 2014 presents much higher mean values for variables X1, X2, X3 and X4, corresponding to the concentrations of PM10, SO2, NO2 and CO. On the other hand, it presents the lowest mean value for variable X5, corresponding to the concentration of O3.

The year 2013 generally presents the lowest mean values for pollutants X1 to X4 and the highest mean value for X5. The year 2015 presents intermediate values between 2013 and 2014, which may make its separation from the other groups more difficult.

4.3. Number of discriminant functions

The maximum number of discriminant functions that can be obtained is given by $m = \min(k - 1, p)$, where k represents the number of groups and p represents the number of explanatory variables.

In this case, there are 3 groups, corresponding to the years A2013, A2014 and A2015, and 5 quantitative variables. Therefore, $m = \min(3 - 1, 5) = \min(2, 5) = 2$. Thus, two linear discriminant functions can be established.

4.4. Linear discriminant functions

The discriminant functions are linear combinations of the original variables. They can be written in the following general form:

$$LD1 = a_{11}X_1 + a_{12}X_2 + a_{13}X_3 + a_{14}X_4 + a_{15}X_5$$

$$LD2 = a_{21}X_1 + a_{22}X_2 + a_{23}X_3 + a_{24}X_4 + a_{25}X_5$$

Variable	LD1	LD2
X1	0.02889	0.01680
X2	0.00812	-0.02827
X3	-0.00944	-0.07193
X4	-0.00032	-0.00069
X5	-0.03650	-0.07931

Table 4 - Coefficients of the two linear discriminant functions.

```
> lda_model <- lda(X6 ~ X1 + X2 + X3 + X4 + X5, data = dados)
> coefficients <- round(lda_model$scaling, 5)
> print(coefficients)
      LD1      LD2
X1  0.02889  0.01680
X2  0.00812 -0.02827
X3 -0.00944 -0.07193
X4 -0.00032 -0.00069
X5 -0.03650 -0.07931
```

Figure 3 - R output showing the coefficients of the discriminant functions.

Using these coefficients, the discriminant functions are approximately:

$$LD1 = 0.02889X_1 + 0.00812X_2 - 0.00944X_3 - 0.00032X_4 - 0.03650X_5$$

$$LD2 = 0.01680X_1 - 0.02827X_2 - 0.07193X_3 - 0.00069X_4 - 0.07931X_5$$

The first discriminant function is mainly influenced by variables X5 and X1. This function is mostly associated with the separation between observations with different levels of O3 and PM10. The second discriminant function has a more relevant contribution from variables X5 and X3, although it explains a much smaller part of the variability between the groups.

It is important to note that the direct interpretation of the coefficients should be made carefully, since the variables are measured on different scales. Therefore, the graphical analysis and the percentage of variability explained by the discriminant functions are especially important for the interpretation of the results.

4.5. Variability explained by the discriminant functions

The eigenvalues associated with the discriminant functions allow us to evaluate the relative importance of each function in separating the groups.

Function	Eigenvalue	Between-group variability explained	Canonical correlation
LD1	181.2926	96.51%	0.9973
LD2	6.5606	3.49%	0.9315

Table 5 - Importance of the discriminant functions.

The first discriminant function explains 96.51% of the between-group variability, while the second function explains only 3.49%. Therefore, almost all the separation between the three years is explained by the first discriminant function.

The canonical correlation of the first function is equal to 0.9973, which is close to 1 and indicates a strong relationship between the first discriminant function and the groups defined by year. On the other hand, the second function has a canonical correlation of 0.9315, indicating a slightly weaker, but still strong, relationship with the groups.

4.6. Global discriminating power

To evaluate the global discriminating power of the functions, Wilks' Lambda test can be used. This test verifies whether the discriminant functions have significant ability to distinguish the groups.

```
> wilks_lambda <- prod(1 / (1 + eigenvalues))
> chi_square <- -((n - 1) - ((p + k) / 2)) * log(wilks_lambda)
> df <- p * (k - 1)
> p_value <- pchisq(chi_square, df = df, lower.tail = FALSE)
> cat("Wilks' Lambda =", round(wilks_lambda, 4), "\n")
Wilks' Lambda = 7e-04
> cat("Chi-square =", round(chi_square, 4), "\n")
Chi-square = 831.285
> cat("df =", df, "\n")
df = 10
> cat("p-value =", format.pval(p_value, digits = 4), "\n")
p-value = < 2.2e-16
```

Figure 4 - R output for Wilks' Lambda test.

The hypotheses considered are:

H0: The discriminant functions do not have discriminating power.

H1: At least one discriminant function has discriminating power.

The test was assessed at a significance level of $\alpha = 0.05$. According to the R output, the p-value was lower than 0.001.

Thus, the null hypothesis is rejected, and it can be concluded that there is sufficient statistical evidence to state that the discriminant functions have significant discriminating power. In other words, variables X1 to X5 allow the three groups defined by year to be distinguished in a statistically significant way.

4.7. Graphical representation of the observations

Next, the graphical representation of the observations in the plane defined by the two discriminant functions was analysed.

	Year	LD1	LD2
1	A2013	-2.9972	-0.3556
2	A2013	-2.2976	0.0101
3	A2013	-2.8160	-0.1674
4	A2013	-2.7795	-0.0881
5	A2013	-2.6980	-0.1573
6	A2013	-2.5129	-0.2146
7	A2013	-2.3165	-0.1500
8	A2013	-2.5279	-0.3148
9	A2013	-2.2651	-0.3039
10	A2013	-2.6674	-0.4967

Figure 5 - R output showing the first ten discriminant scores.

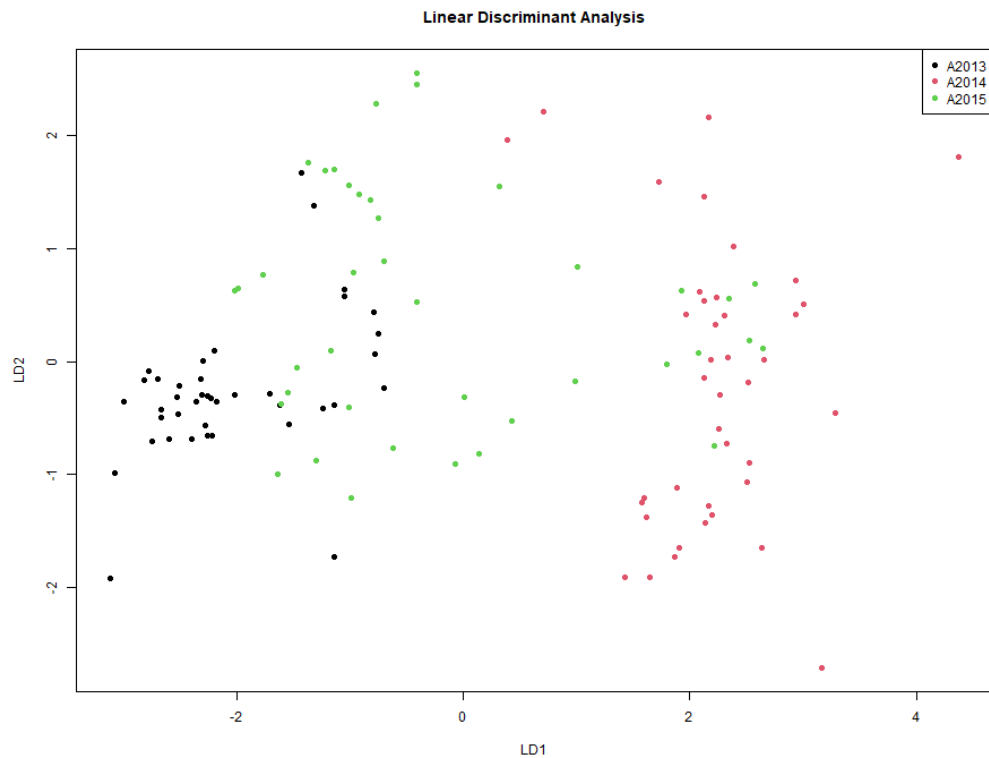


Figure 6 - Graphical representation of the observations on LD1 and LD2.

From the observation of Figure 6, it can be seen that the group corresponding to the year 2014 is relatively well separated from the other groups, mainly along the first discriminant function. This result is consistent with the analysis of the group means, since the year 2014 presents higher values for variables X1, X2, X3 and X4, and lower values for X5.

On the other hand, the groups corresponding to the years 2013 and 2015 are closer to each other. Group A2015 occupies an intermediate position between A2013 and A2014, which indicates that its atmospheric pollution profile is not as clearly distinct. This overlap helps to explain the classification errors observed later.

4.8. Classification matrix

In order to evaluate the effectiveness of the linear discriminant analysis, the classification matrix obtained by the resubstitution method was constructed.

Original	Predicted		
	A2013	A2014	A2015
A2013	32	0	8
A2014	0	38	2
A2015	11	9	20

Figure 7 - R output with the classification matrix obtained by the resubstitution method.

Original group / Predicted group	A2013	A2014	A2015
A2013	32	0	8
A2014	0	38	2
A2015	11	9	20

Table 6 - Classification matrix obtained by the resubstitution method.

From the classification matrix, it can be seen that 32 observations from group A2013 were correctly classified, while 8 were classified as A2015. In the case of group A2014, 38 observations were correctly classified and only 2 were classified as A2015. Finally, in group A2015, only 20 observations were correctly classified, while 11 were classified as A2013 and 9 as A2014.

The percentage of correct classification in each group is given by n_{ii}/n_i . The following results were obtained:

Group	Correctly classified	Correct classification	Incorrect classification
A2013	32/40	80.00%	20.00%
A2014	38/40	95.00%	5.00%
A2015	20/40	50.00%	50.00%

Table 7 - Correct and incorrect classification percentages by group.

Group	Correctly_classified	Correct_classification	Incorrect_classification
1 A2013	32/40	80%	20%
2 A2014	38/40	95%	5%
3 A2015	20/40	50%	50%

Figure 8 - R output with the correct and incorrect classification percentages by group.

The total percentage of correct classification is given by $(32 + 38 + 20) / 120 = 90 / 120 = 0.75$. Thus, the total percentage of correct classification is 75.00%.

These results show that group A2014 is the best classified group, with 95.00% of the observations correctly assigned. This confirms that the atmospheric pollution profile of 2014 is quite different from the other years.

Group A2013 also presents a reasonable classification result, with 80.00% of correct classifications. However, some observations from this group were classified as belonging to A2015, indicating some similarity between these two years.

Group A2015 presents the weakest performance, with only 50.00% of correct classifications. This result suggests that the 2015 profile is located in an intermediate position between 2013 and 2014, making its clear separation from the other groups more difficult.

4.9. Cross-validation

To complement the evaluation of the model, cross-validation using the leave-one-out method was also considered. This procedure consists of removing one observation from the sample, fitting the model with the remaining observations, and then classifying the removed observation. The process is repeated for all observations.

	Predicted		
Original	A2013	A2014	A2015
A2013	32	0	8
A2014	0	38	2
A2015	11	9	20

Figure 9 - R output with the classification matrix obtained by leave-one-out cross-validation.

Original group / Predicted group	A2013	A2014	A2015
A2013	32	0	8
A2014	0	38	2
A2015	11	9	20

Table 8 - Classification matrix obtained by leave-one-out cross-validation.

The cross-validation produced the same total percentage of correct classification, equal to 75.00%. This result suggests that the performance of the model is relatively stable.

However, as in the resubstitution method, group A2015 remains the most difficult group to classify. Therefore, the main limitation of the linear discriminant analysis is the separation of the year 2015 from the years 2013 and 2014.

4.10. Conclusion of the linear discriminant analysis

The linear discriminant analysis allowed us to evaluate whether the concentrations of atmospheric pollutants make it possible to distinguish the observations corresponding to the years 2013, 2014 and 2015.

The results obtained show that variables X1 to X5 have significant discriminating power. The first discriminant function explains 96.51% of the between-group variability, indicating that the separation between the years is mainly achieved through one dominant discriminant direction.

The year 2014 was the easiest to distinguish, since it presents higher mean values for variables X1, X2, X3 and X4 and lower values for X5. This behaviour resulted in most observations from 2014 being correctly classified.

On the other hand, the year 2015 was more difficult to classify. This is due to the fact that its pollution profile is intermediate between the years 2013 and 2014, causing some overlap with both groups.

Overall, the linear discriminant analysis achieved a total correct classification rate of 75.00%, which indicates a reasonable ability to discriminate between the three years. Therefore, it can be concluded that the method allows relevant differences in atmospheric pollution profiles to be identified, although the separation between all groups is not perfect.

```
> cat("Best classified group:", names(which.max(correct_by_group)), "\n")
Best classified group: A2014
> cat("Weakest classified group:", names(which.min(correct_by_group)), "\n")
Weakest classified group: A2015
> cat("Overall accuracy:", paste0(round(accuracy * 100, 2), "%"), "\n")
Overall accuracy: 75%
> cat("LD1 explains:", paste0(round(prop_variability[1] * 100, 2), "%"), "\n")
LD1 explains: 96.51%
> cat("LD2 explains:", paste0(round(prop_variability[2] * 100, 2), "%"), "\n")
LD2 explains: 3.49%
```

Figure 10 - R output summarising the main conclusions of the LDA classification.